WattIf

An AI-assisted clean-energy planning simulator for equitable renewable siting.

Sardul · Sujal · Daniel · Sakib

TEAM VELOCITY

WHY NOW

Clean-energy planning is bottlenecked before shovels hit the ground.

WattIf compresses the early planning loop: match siting potential with demand, equity, community risk, and grid capacity before a proposal turns into sunk cost.

55 mo.

typical grid-connection timeline

The typical project built in 2024 took 55 months from interconnection request to commercial operation.

Berkeley Lab, Queued Up 2025

13%

of queued capacity got built

Only 13% of capacity requesting interconnection from 2000-2019 had reached operation by the end of 2024.

Berkeley Lab, Queued Up 2025

~50%

face 6+ month siting delays

About half of recent utility-scale wind and solar siting applications experienced delays of six months or more.

Berkeley Lab developer survey, 2024

55%

of Toronto emissions from buildings

Buildings remain Toronto's largest emissions source, so local clean energy planning has to match demand, equity, and grid constraints.

City of Toronto SBEI, 2023

CHALLENGE SET 1 · THEME 1

The problem we're targeting

1

Renewable Siting & Demand Matching

Identify where renewable installations will be most effective by combining energy potential and local demand.

2

DER Capacity & Grid Stability

Understand how much DER capacity, including solar, EVs, and batteries, the grid can support without overload.

Product demo

Tech stack · Features · Highlights

TECH STACK

- React 19, Vite, TypeScript, Tailwind, Zustand
- Mapbox / MapLibre + deck.gl 3D planning map
- FastAPI simulation backend + WebSocket live ticks
- Supabase Postgres for proposals, placements, snapshots
- Anthropic / Featherless planner with scripted fallback

FEATURES

- 140 Toronto neighbourhoods with demand, equity, risk, constraints, and clean-asset overlays
- Manual, optimizer, or AI-assisted placement for solar, wind, batteries, microgrids, and EV chargers
- Monthly metrics: coverage, equity, approval, emissions, grid load, cost
- 16+ stress scenarios including blackout, heatwave, ice storm, flood, EV surge, and gas spike
- Snapshots, comparisons, and exportable decision-support memo

DIFFERENTIATORS

- Equity-weighted optimizer ranks by renewable potential and energy burden, not cost alone
- Real Toronto data lane: open data, Census signals, OSM buildings, PVGIS solar, city clean assets
- ~8,000 simulated residents and template voices for zone-by-zone sentiment
- Honesty badges show Live vs Mock API, LLM vs Demo planner, Supabase vs in-memory
- Hackathon-safe offline mode when backend, keys, or network are unavailable

VELOCITY · SENECA ENERGY HACKATHON

Thank you.

We appreciate the opportunity to share WattIf and the future we imagine for faster, fairer clean-energy planning.

github.com/DanielWLiu07/wattif